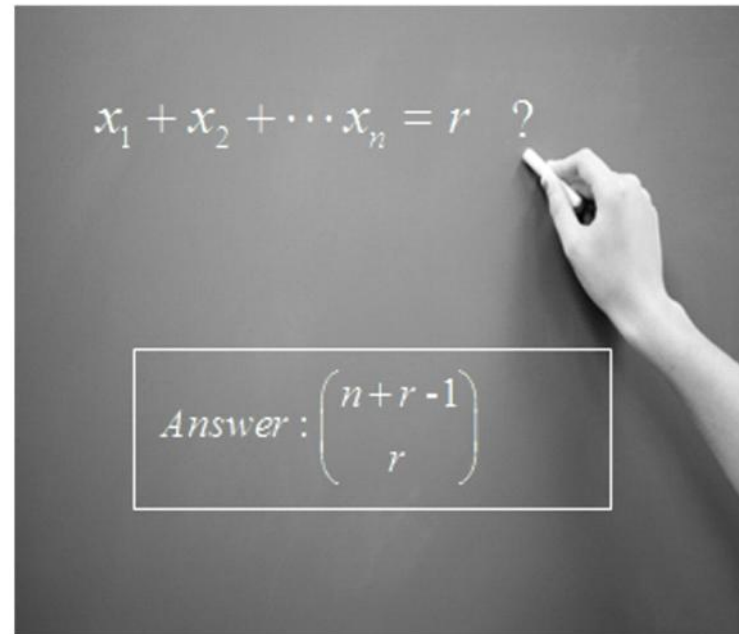

COMP 2121
DISCRETE
MATHEMATICS

Lecture 4



Combinations with Repetition Allowed

Nested Loops

1 2 3 4 5

Example 1. (Review problem): Consider the following program segments. What value does a variable `counter` have after the following code execution?

```
counter = 15
for i = 1 to 5
  for j = 1 to 5
    counter = counter + 8
  next j
  counter = counter + 9
next i
```

$$\text{counter} = 15 + \frac{(5)(5)}{i, j} (8) + \frac{(5)}{i} (9)$$

1 2 3 4 5

```
counter = 15
for i = 1 to 5
  for j = i+1 to 5
    counter = counter + 8
  next j
  counter = counter + 9
next i
```

$$\text{counter} = 15 + \frac{\binom{5}{2}}{i, j} (8) + \frac{(5)}{i} (9)$$

```
counter = 15
for i = 1 to 6
  for j = i+1 to 10
    counter = counter + 8
  next j
  counter = counter + 9
next i
```

$$\text{counter} = 15 + \frac{\left[\binom{10}{2} - \binom{4}{2} \right]}{i, j} (8) + \frac{(6)}{i} (9)$$

1 2	2 3	3 4	4 5	5 6	6 7
1 2	2 4	3 5	4 6	5 7	6 8
1 4	:	:	:	:	6 9
:	:	3 10	4 10	5 10	6 10
:	2 10				
1 10					

7 8 8 9 9 10
7 9 8 10
7 10
7, 8, 9, 10 $\binom{4}{2}$

$$\leftarrow \binom{10}{2} \rightarrow$$

Example 2. In how many ways can one distribute 12 identical blue marbles among three math instructors: Andrew, David and Scott?

AAAA DDDD SSSS $\leftrightarrow$ AADDAA DDSSSS

AAAA AAAA AAAA
AAAA AAAA A SSS

$\leftarrow \begin{array}{c} \leftarrow \\ A \end{array} \begin{array}{c} \rightarrow \\ D \end{array} \begin{array}{c} \rightarrow \\ S \end{array}$

XXXX | XXX | XXXX

XXXX XXXX XXXX |

XXXX XXXX X|XXX

How many different strings of 12 x's & 2 1's

do we have? $\frac{14!}{12!2!}$

-----1-----1-----

$$\binom{14}{2} = \binom{14}{12}$$

$n=3$ $r=12$

$$x_1 + x_2 + x_3 = 12$$

$x_1 \geq 0$ $x_2 \geq 0$ $x_3 \geq 0$

$$\binom{r + (n-1)}{n-1} = \binom{r + (n-1)}{r}$$

$$\binom{n+r-1}{r}$$

$$A + D + S = 12$$

$$A \geq 0 ; D \geq 0 ; S \geq 0$$

$$A + D + S = 12$$

$$A \geq 0 ; D \geq 0 ; S \geq 0$$

$$\begin{array}{ccccccc} A & D & S & & & & \\ \cancel{A} & \cancel{D} & \cancel{S} & 0 & 0 & & \\ & 0 & 0 & 0 & 0 & 0 & \end{array}$$

$$n=3$$

$$r=12$$

$$\binom{n+r-1}{r}$$

$$= \binom{3+12-1}{12}$$

$$= \binom{14}{12}$$

$$n=3$$

$$r=9$$

$$\binom{n+r-1}{r} = \binom{3+9-1}{9} = \binom{11}{9}$$

$$9 \quad 0 \quad 7 \quad 9 \quad 9 \quad 9$$

$$\begin{pmatrix} n+1-1 \\ 9 \end{pmatrix} = \begin{pmatrix} 3+4-1 \\ 9 \end{pmatrix} = \begin{pmatrix} 6 \\ 9 \end{pmatrix}$$

Formal Definition

Theorem: When we wish to select, with repetition, r of n distinct objects, we find that we are considering all arrangements of r 'x's and $n-1$ |'s and thus their number is:

$$\binom{n+r-1}{r} = \frac{(n+r-1)!}{r!(n-1)!}$$

All combinations with repetition allowed
of 1, 2, 3, 4 size 2

$$n=4; r=2$$

1 1 ✓ 2 2 ✓ 3 3 ✓ 4 4 ✓
1 2 ✓ 2 3 ✓ 3 4 ✓
1 3 ✓ 2 4 ✓
1 4 ✓

$$\binom{n+r-1}{r} = \binom{4+2-1}{2}$$

$$= \binom{5}{2} = \frac{5 \cdot 4}{1 \cdot 2} = 10$$

"CANADIAN" - problem revisited

Example 3. How many arrangements of the letters in CANADIAN have at least two letters between the N's? (Similar to example from Lecture 3)

Handwritten notes illustrating the solution to the problem:

Initial arrangement: C A A A D I N N

Step 1: Treat the two N's as a single unit (N N). The remaining letters are C, A, A, A, D, I. The number of arrangements of these 6 letters is $\frac{6!}{3!}$.

Step 2: Consider the positions of the N's. The N's can be placed in the gaps between the 6 letters, resulting in 4 possible positions for the first N and 3 possible positions for the second N. The number of ways to choose 2 positions out of 4 is $\binom{4}{2}$.

Step 3: The total number of arrangements is the product of the number of ways to place the N's and the number of arrangements of the other letters: $\binom{4}{2} \cdot \frac{6!}{3!}$.

Final answer: $\binom{6}{4}$.

Alternative approach using stars and bars:

Let x_1 be the number of letters between the two N's, x_2 be the number of letters before the first N, and x_3 be the number of letters after the second N. We have $x_1 + x_2 + x_3 = 4$ (since there are 4 gaps between the 6 letters). The number of solutions to this equation is $\binom{n+r-1}{r} = \binom{3+4-1}{4} = \binom{6}{4}$.

Final calculation: $\binom{6}{4} \cdot \frac{6!}{3!}$.

Example 4. Consider the following program segments. How many times is the print statement executed?

```
for i = 1 to 5
  for j = i to 5
    print i, j
  next j
next i
```

```
for i = 1 to 5
  for j = 1 to i
    print i, j
  next j
next i
```

SEE LABS FOR LOOP PROBLEMS

Summary:



	Order matters	Order does not matter
Repetition is allowed	n^k	$\binom{n+k-1}{k}$
Repetition is Not allowed	$P(n, k) = \frac{n!}{(n-k)!}$	$\binom{n}{k} = \frac{n!}{k!(n-k)!}$

+ MISSISSIPPI